



COURSE SLIDES · WSQ

SQL Fundamental for Beginners

WSQ Course Code: TGS-2020505790

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for WSQ-funded courses.

2

QR code from SSG

The trainer or administrator displays the attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum 75% attendance rate is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer & Founder, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS

PhD — over 20 years of training and consulting experience in data, analytics and software engineering.

DELIVERS

WSQ courses on SQL, databases, data engineering, analytics and AI.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1

Your name, organisation and role

2

Your experience with spreadsheets,
databases or programming

3

The data you want to query and
analyse after this course

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

5

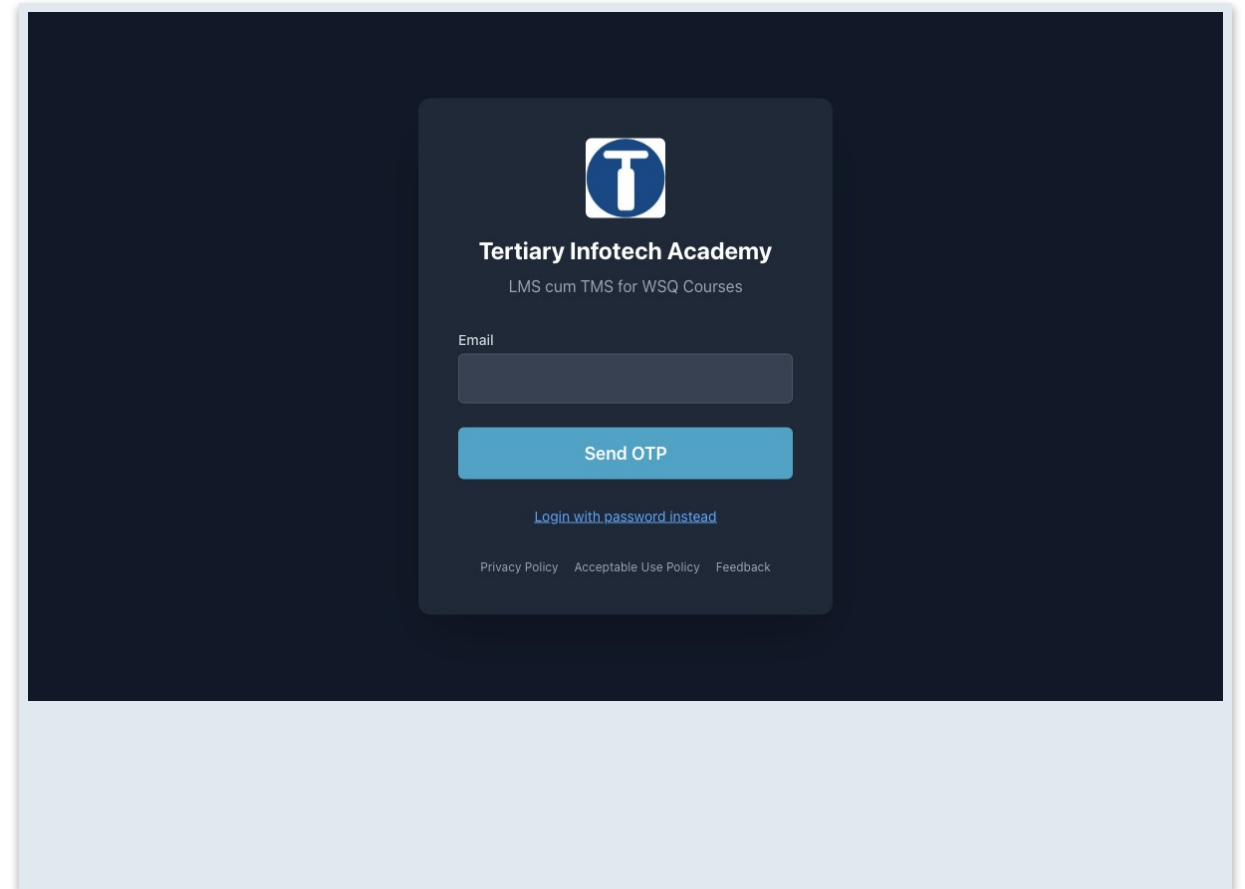
Be punctual; return from breaks on time.

6

75% attendance is required.

Download Course Material

- 1 Log in to the LMS portal: lms-tms.tertiaryinfotech.com
- 2 Open My Courses and select SQL Fundamental for Beginners.
- 3 Open the Course Materials section of the course.
- 4 Download the slides, Learner Guide and world.db sample database.



SCHEDULE

Lesson Plan — 1 Day, 8 Training Hours

Morning

- Morning (9:00 am – 1:00 pm)
 - Welcome, introductions & digital attendance (AM)
 - Topic 1: Data Modeling — Labs 1–3
 - Topic 2: Data Processing & Analysis — Labs 4–6
 - Lunch break (1:00 – 2:00 pm)

Afternoon

- Afternoon (2:00 pm – 6:00 pm)
 - Topic 3: Data Transformation — Labs 7–9
 - Topic 4: Introduction to Data Warehouse — Labs 10–11
 - Course feedback (TRAQOM) & Assessment digital attendance
 - Briefing, then Final Assessment: PP (70 min) + OQ (20 min)

Skills Framework

- TSC Title: Data Engineering
 - TSC Code: ICT-DIT-3005-1.1
 - Framework: ICT Skills Framework
- This course develops the TSC abilities:
 - A1: Identify relevant data sources, processes and relationships in accordance to business requirements
 - A2: Propose methods and tools to gather and process data, and minimise confounding variables and data limitations
 - A3: Apply data analysis and data profiling to improve the clarity, quality and integrity of valid data
 - A4: Process multiple streams of data using data systems
 - A5: Utilise data systems and platform capabilities to solve new data problems

Skills Framework — TSC Abilities (continued)

- A6: Transform data to meet specified business requirements
- A7: Operate data warehouse systems to balance optimisation of data access with batch loading and resource utilisation
- A8: Perform data mapping from source to destination systems
- A9: Resolve data issues arising from data processing activities
- Knowledge items K1–K7 (data sources, modeling, warehousing, queries, transformation, mapping) are covered across Topics 1–4.

Learning Outcomes

1

LO1 · Data Modeling

Apply data modeling for business processes.

2

LO2 · Data Processing

Apply data processing and analysis using SQL.

3

LO3 · Data Transformation

Apply data transformation from multiple data sources.

4

LO4 · Data Warehouse

Apply data mapping to data warehouse.

Course Outline

1

Topic 1 · Data Modeling (A1, A2, K1, K2)

Data Sources · Data Modeling · SQL Data Types · Create & Manage Databases

2

Topic 2 · Data Processing and Analysis (A3, A5, K4)

SQL Queries · SQL Operators · Insert, Update & Delete

3

Topic 3 · Data Transformation (A4, A6, K5, K6)

Aggregate Data · Join Data · Group Data

4

Topic 4 · Introduction to Data Warehouse (A7, A8, A9, K3, K7)

Overview of Data Warehouse · Relational Databases · Stored Procedures

BEFORE YOU BEGIN

Briefing for Assessment

1

Place phones and other materials under the table or on the floor.

2

No photos or recording of assessment scripts.

3

No discussion during the assessment.

4

Use a black or blue pen for hard-copy assessments.

5

No liquid paper or correction tape.

6

Assessment scripts are collected when the time is up.

Final Assessment

1

Practical Performance (PP)

Hands-on SQL tasks in SQLite Studio · 70 minutes
· assessor-to-candidate ratio 1:10.

2

Oral Questioning (OQ)

Spoken knowledge questions · 20 minutes · one-to-one with the assessor.

3

Open book

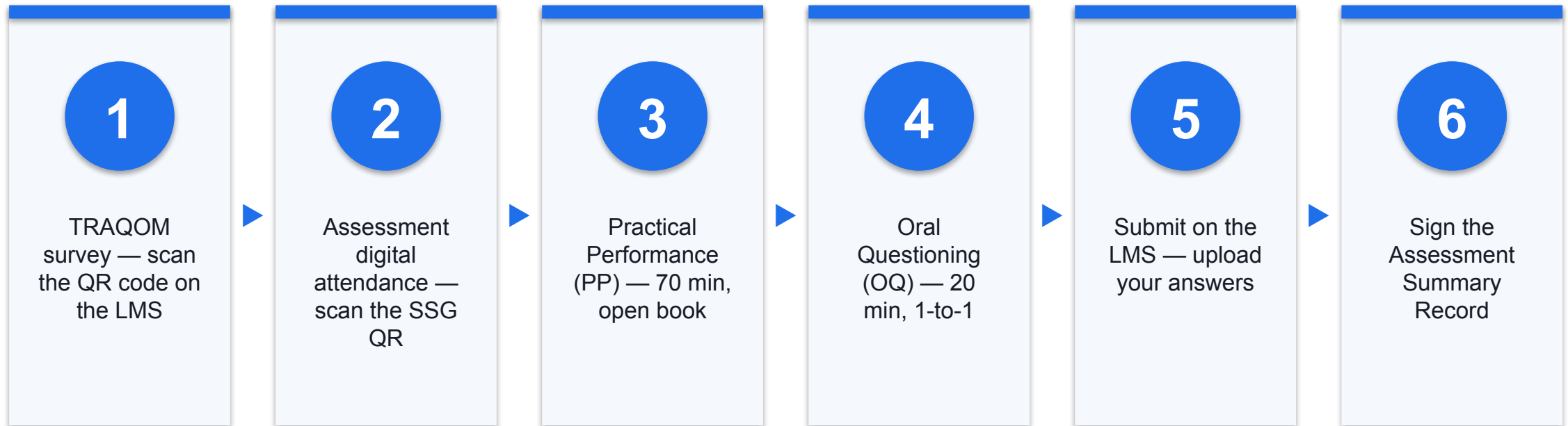
Course slides, Learner Guide and approved materials only.

4

To be funded

75% minimum attendance and a result of 'Competent'. An appeal process is available.

Assessment Flow





SQL FOUNDATIONS

What is SQL?

What is SQL?

1

Structured Query Language

The standard language for storing, manipulating and retrieving data in databases.

2

An open standard

ANSI standard since 1986, ISO standard since 1987.

3

Works everywhere

MySQL, SQL Server, Access, Oracle, Sybase, Informix, PostgreSQL and more.

4

Declarative

You state WHAT data you want; the database engine works out how to get it.

What Can SQL Do?

1

Query

Execute queries and retrieve data from a database.

2

Change data

Insert, update and delete records.

3

Define structure

Create new databases, tables, views and indexes.

4

Automate

Create stored procedures for reusable server-side logic.

5

Secure

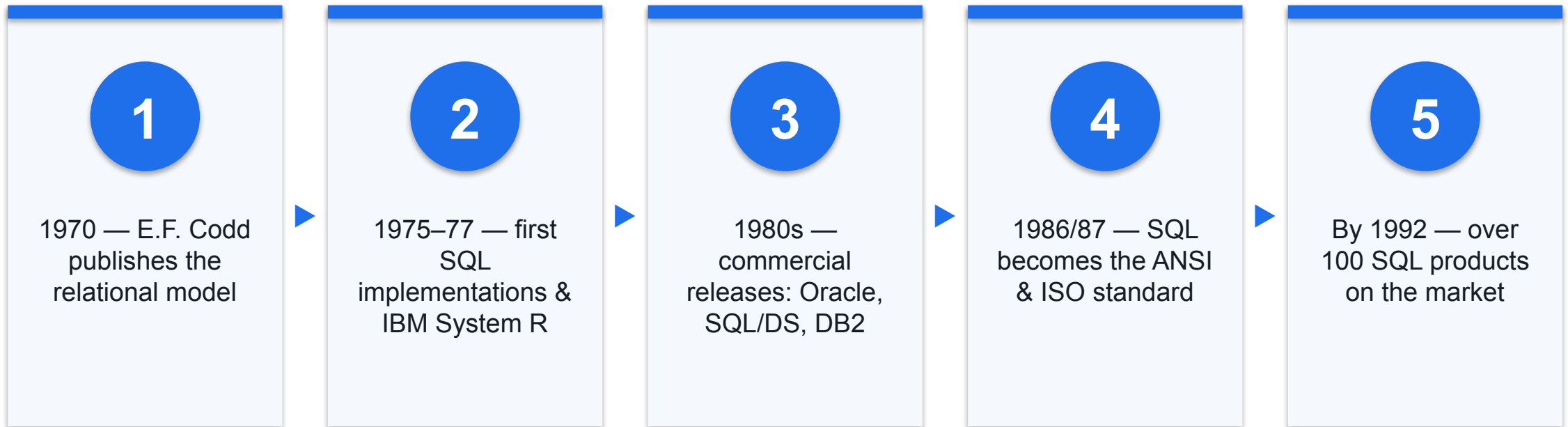
Set permissions on tables, procedures and views.

6

Analyse

Aggregate, join and group data for analysis.

A Short History of SQL



Databases That Speak SQL

1

MySQL / MariaDB

The web's workhorse open-source databases.

2

PostgreSQL

Advanced open-source object-relational database.

3

SQLite

Zero-config embedded database — our classroom engine.

4

Microsoft SQL Server

Enterprise database with T-SQL and stored procedures.

5

Oracle

Enterprise market leader since 1980.

6

DB2 · Sybase · Access

IBM's DB2, SAP's Sybase and desktop MS Access.

Components of a Table

Rows (records)

- A database is one or more tables
 - Each table is a collection of rows
 - Every row of a table has the same row type
 - A row is the smallest unit inserted or deleted

Columns (fields)

- Columns define the fields
 - The n-th field of every row is the n-th column
 - Each column has a name and a data type
 - Rows × columns = the relational grid

Your Workbench — SQLite Studio

1

Desktop app

Browse and edit SQLite database files visually — created by Paweł Salawa.

2

Open source

Free, GPLv3-licensed, cross-platform (C++/Qt) — download at sqlitestudio.pl.

3

SQL editor built in

Write and run every query in this course from Tools → Open SQL editor.

4

Cloud alternative

No install? Use the free sqliteonline.com in your browser.

TOPIC 01

Data Modeling

Data Sources · Data Modeling · SQL Data Types · Create & Manage Databases

Key Concepts — Data Modeling

1

Relational databases

A database is one or more tables; each table is rows (records) and columns (fields) with a fixed row type.

2

DDL statements

CREATE / ALTER / DROP define databases, tables, indexes and their columns and data types.

3

Constraints

NOT NULL, DEFAULT, UNIQUE, CHECK, PRIMARY KEY and FOREIGN KEY enforce rules on the data.

4

Data modeling

Primary and foreign keys link tables into entity relationships that mirror the business process.

CREATE DATABASE Statement

- The CREATE DATABASE statement creates a new SQL database.
- DROP DATABASE permanently deletes an existing database — use with care.

SYNTAX

```
CREATE DATABASE databasename;
```

```
DROP DATABASE databasename;
```

EXAMPLE

```
CREATE DATABASE testDB;
```

```
DROP DATABASE testDB;
```

CREATE TABLE Statement

- CREATE TABLE creates a new table in a database.
- Each column is declared with a name and a data type.

SYNTAX

```
CREATE TABLE table_name (  
    column1 datatype,  
    column2 datatype,  
    column3 datatype  
);
```

EXAMPLE

```
CREATE TABLE Persons (  
    PersonID int,  
    LastName varchar(255),  
    FirstName varchar(255),  
    City varchar(255)  
);
```

String Data Types

1

CHAR(n)

Fixed-length string, padded to n characters (0–255).

2

VARCHAR(n)

Variable-length string up to n characters — the everyday text type.

3

TEXT

Long free text (articles, descriptions) up to 65,535 bytes.

4

BLOB

Binary large object — images, files and other raw bytes.

5

ENUM

One value chosen from a fixed list you define.

6

SET

Zero or more values chosen from a fixed list.

Numeric Data Types

1

INT / INTEGER

Whole numbers — IDs, counts, populations.

2

SMALLINT / BIGINT

Smaller or larger integer ranges to fit the data.

3

DECIMAL(p,s)

Exact fixed-point numbers — money and precise quantities.

4

FLOAT / REAL / DOUBLE

Approximate floating-point numbers for scientific values.

5

BOOLEAN

True/false flags (stored as 0 / 1).

6

BIT

Individual bit values for compact flags.

Date and Time Data Types

1

DATE

Calendar date: YYYY-MM-DD.

2

TIME

Time of day: hh:mm:ss.

3

DATETIME

Date and time combined: YYYY-MM-DD hh:mm:ss.

4

TIMESTAMP

Date-time stored relative to epoch — auto-updates on change.

5

YEAR

Four-digit year.

6

Note

SQLite stores dates as TEXT, REAL or INTEGER under the hood.

ALTER TABLE Statement

- ALTER TABLE adds, deletes or modifies columns in an existing table.
- It is also used to add or drop constraints after creation.

SYNTAX

```
ALTER TABLE table_name  
ADD column_name datatype;
```

EXAMPLE

```
ALTER TABLE Customers  
ADD Email varchar(255);
```

SQL Constraints

1

NOT NULL

Column must always contain a value.

2

UNIQUE

All values in the column must be different.

3

PRIMARY KEY

NOT NULL + UNIQUE — uniquely identifies each row; one per table.

4

FOREIGN KEY

Links a column to the primary key of another table.

5

CHECK

Limits the values allowed in a column by a condition.

6

DEFAULT

Supplies a value automatically when none is given.

NOT NULL & DEFAULT Constraints

- NOT NULL forces a column to always hold a value.
- DEFAULT fills in a value automatically when the INSERT omits one.

SYNTAX

```
CREATE TABLE t (  
    col datatype NOT NULL,  
    col2 datatype DEFAULT 'value'  
);
```

EXAMPLE

```
CREATE TABLE Persons (  
    ID int NOT NULL,  
    LastName varchar(255) NOT NULL,  
    City varchar(255) DEFAULT 'Sandnes'  
);
```

UNIQUE & CHECK Constraints

- UNIQUE guarantees no duplicate values in a column — a table may have many UNIQUE constraints.
- CHECK limits the range of values a column accepts.

SYNTAX

```
CREATE TABLE t (  
    col int NOT NULL UNIQUE,  
    age int CHECK (age >= 18)  
);
```

EXAMPLE

```
CREATE TABLE Persons (  
    ID int NOT NULL UNIQUE,  
    Age int CHECK (Age >= 18)  
);
```

PRIMARY KEY Constraint

- The PRIMARY KEY uniquely identifies each record — UNIQUE and NOT NULL combined.
- A table has exactly ONE primary key; it may span one or several columns.

SYNTAX

```
CREATE TABLE t (  
    ID int NOT NULL,  
    ...,  
    PRIMARY KEY (ID)  
);
```

EXAMPLE

```
CREATE TABLE Persons (  
    ID int NOT NULL,  
    LastName varchar(255) NOT NULL,  
    PRIMARY KEY (ID)  
);
```

FOREIGN KEY Constraint

- A FOREIGN KEY links two tables: a field in the child table refers to the PRIMARY KEY of the parent table.
- This is the mechanism that turns separate tables into a data model.

SYNTAX

```
CREATE TABLE child (  
    ...,  
    FOREIGN KEY (col)  
    REFERENCES parent(pk_col)  
);
```

EXAMPLE

```
CREATE TABLE Orders (  
    OrderID int NOT NULL,  
    PersonID int,  
    PRIMARY KEY (OrderID),  
    FOREIGN KEY (PersonID)  
    REFERENCES Persons(PersonID)  
);
```

AUTO INCREMENT Field

- Auto-increment generates a unique number automatically for each new record.
- Usually applied to the primary key. In SQLite an INTEGER PRIMARY KEY auto-increments by default.

SYNTAX

```
CREATE TABLE t (  
    id int NOT NULL AUTO_INCREMENT,  
    ...,  
    PRIMARY KEY (id)  
);
```

EXAMPLE

```
CREATE TABLE Persons (  
    Personid int NOT NULL AUTO_INCREMENT,  
    LastName varchar(255) NOT NULL,  
    PRIMARY KEY (Personid)  
);
```

CREATE INDEX Statement

- Indexes speed up searches and queries — the database finds rows without scanning the whole table.
- Users never see the index; it works silently behind SELECT and WHERE.

SYNTAX

```
CREATE INDEX index_name  
ON table_name (column1, column2);
```

EXAMPLE

```
CREATE INDEX idx_lastname  
ON Persons (LastName);
```

Data Modeling with Keys

What data modeling is

- Explores data-oriented structures for the business
 - Identifies entity types (like classes) and their attributes
 - Spans conceptual → logical → physical models
 - Mirrors real business processes and relationships

How keys create the model

- PRIMARY KEY uniquely identifies each entity row
 - FOREIGN KEY points a child row at its parent
 - One-to-many: one Country has many Cities
 - Relationships form the entity-relationship diagram

SECURITY AWARENESS

Never paste user input straight into SQL.

SQL injection places malicious code in statements via web input — one of the most common hacking techniques. Always validate input and use parameterised queries.

Hands-On Labs — Data Modeling

Lab 1

- Set Up SQLite Studio & Import the World Database

Lab 2

- Create a Database and Tables

Lab 3

- Model Data with Constraints and Keys

Set Up SQLite Studio & Import the World Database

LAB 1

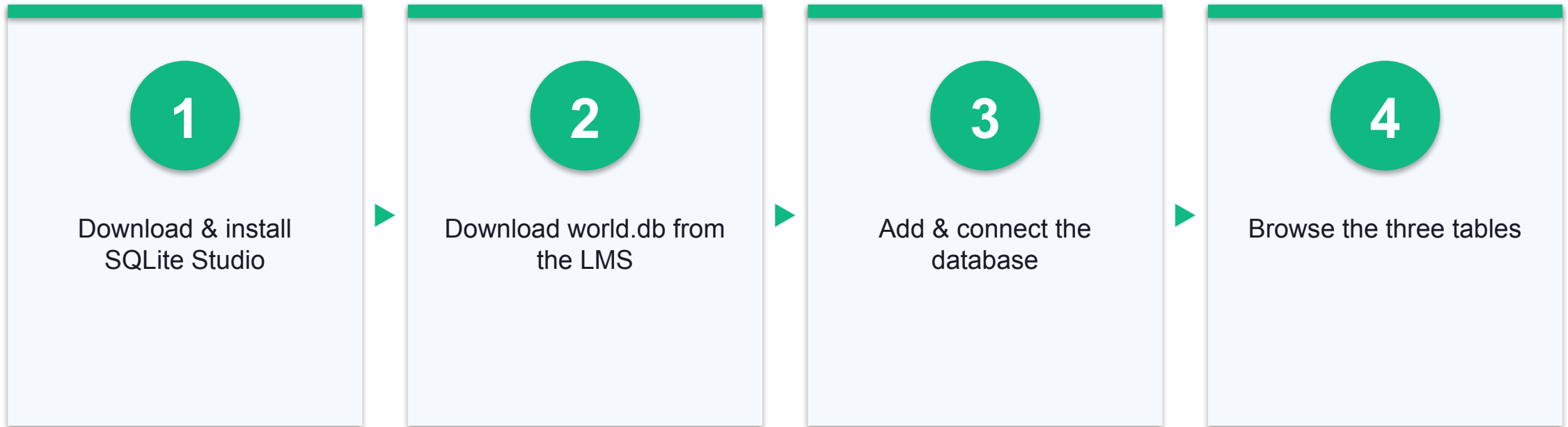
Install SQLite Studio, connect it to the world sample database and explore its tables — your data source for the whole course.

You'll build

A working SQLite Studio with the world database connected, showing the City, Country and CountryLanguage tables.

Tools: SQLite Studio, world.db sample database, LMS course materials

Lab 1 — How It Flows



Set Up SQLite Studio & Import the World Database

Test it

The Databases panel shows the connected world database with the City, Country and CountryLanguage tables, and the Data tab displays city rows (ID, Name, CountryCode, District, Population).

Create a Database and Tables

LAB 2

Create your own database named 'sample' and build the City, Country and CountryLanguage tables with CREATE TABLE, choosing an appropriate data type for every column.

You'll build

A new 'sample' database containing three empty tables whose columns, data types and defaults mirror the world schema.

Tools: SQLite Studio, SQL Editor

Lab 2 — How It Flows



Create a Database and Tables

Test it

The sample database lists City, Country and CountryLanguage; each table's Structure tab shows the declared columns, data types, defaults and primary keys.

Model Data with Constraints and Keys

LAB 3

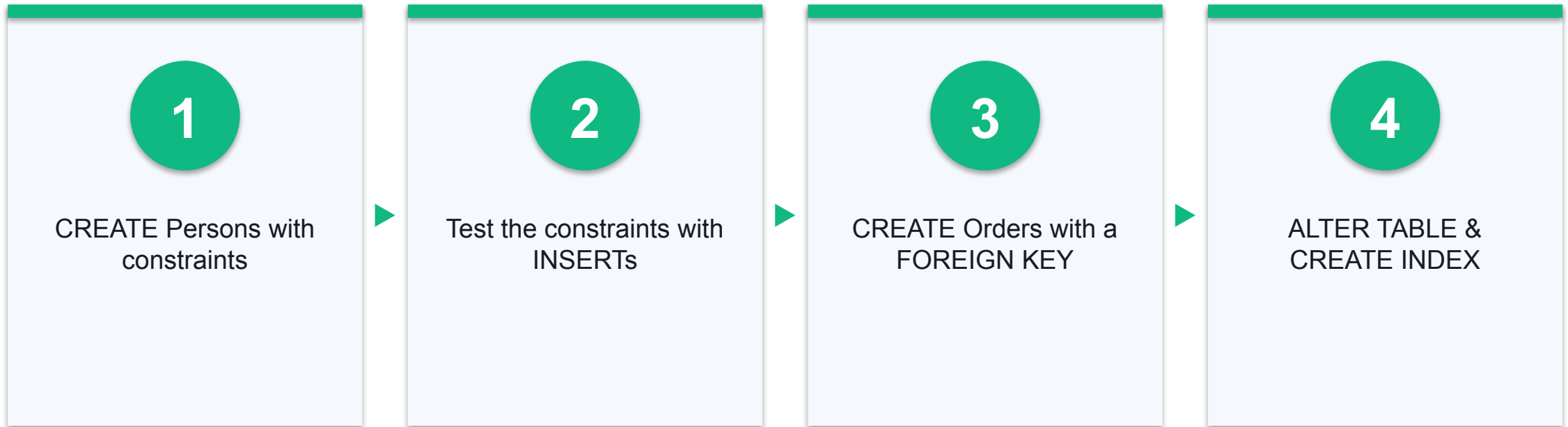
Build a Persons–Orders mini-model: enforce data rules with NOT NULL, UNIQUE, DEFAULT and CHECK, link the tables with a PRIMARY KEY / FOREIGN KEY relationship, and speed up searches with an index.

You'll build

A two-table data model (Persons ← Orders) with working constraints, a foreign-key relationship and an index.

Tools: SQLite Studio, SQL Editor

Lab 3 — How It Flows



Model Data with Constraints and Keys

Test it

The valid insert appears in Persons with City = 'Singapore'; the three rule-breaking inserts each raise a constraint error; Orders shows a foreign key to Persons in its DDL; and idx_lastname is listed under Indexes.

Recap — Data Modeling

- You can now: LO1 — identify relevant data sources and set up the SQL workbench (A1, A2)
- You can now: LO1 — create and manage databases and tables with SQL DDL (A1, A2)
- You can now: LO1 — apply data modeling for business processes with constraints, keys and relationships (A1, A2)
- Full step-by-step instructions for every lab are in the Learner Guide.

Tea Break

15 minutes

TOPIC 02

Data Processing and Analysis

SQL Queries · SQL Operators · Insert, Update & Delete

Key Concepts — Data Processing and Analysis

1

Queries

SELECT retrieves data into a result-set; DISTINCT removes duplicates; WHERE filters rows.

2

Operators

AND/OR/NOT, IN, BETWEEN, LIKE wildcards and IS NULL build precise filter conditions.

3

Shaping results

ORDER BY sorts, LIMIT caps the row count, and aliases (AS) rename columns and tables.

4

Changing data

INSERT INTO adds records, UPDATE modifies them and DELETE removes them — always with WHERE.

Key Elements of SQL

1

Queries

Retrieve data against some criteria — the SELECT family.

2

Statements

Control transactions, sessions, connections and program flow.

3

Clauses

Components of queries and statements — WHERE, GROUP BY, ORDER BY.

4

Expressions

Combinations of symbols and operators producing values.

5

Predicates

Conditions that evaluate true or false to filter rows.

6

Result-set

The table of rows a query returns.

SELECT Statement

- SELECT retrieves data from a database into a result-set.
- Use * for all columns, or name the columns you need.

SYNTAX

```
SELECT column1, column2  
FROM table_name;  
  
SELECT * FROM table_name;
```

EXAMPLE

```
SELECT * FROM City;  
  
SELECT first_name, last_name  
FROM people;
```

SELECT DISTINCT Statement

- SELECT DISTINCT returns only the different (distinct) values.
- Use it when a column holds many duplicates and you want each value once.

SYNTAX

```
SELECT DISTINCT column1, column2  
FROM table_name;
```

EXAMPLE

```
SELECT DISTINCT CountryCode  
FROM City;
```

WHERE Clause

- WHERE filters records — only rows that satisfy the condition are returned.
- Works with =, <, >, <=, >=, <> and the SQL operators.

SYNTAX

```
SELECT column1, column2  
FROM table_name  
WHERE condition;
```

EXAMPLE

```
SELECT * FROM City  
WHERE District = 'Kabul';
```

AND · OR · NOT Operators

1

AND

Row shown only if ALL conditions are true.

2

OR

Row shown if ANY condition is true.

3

NOT

Row shown if the condition is NOT true.

4

Combine freely

Group with parentheses:
WHERE a AND (b OR c).

5

Example

```
WHERE Country='Germany'  
AND City='Berlin'
```

6

Example

```
WHERE NOT Country='UK'
```

IN & BETWEEN Operators

- IN matches any value in a list — shorthand for multiple ORs.
- BETWEEN selects values inside an inclusive range of numbers, text or dates.

SYNTAX

```
WHERE column IN (v1, v2, ...);
```

```
WHERE column BETWEEN v1 AND v2;
```

EXAMPLE

```
SELECT * FROM City  
WHERE CountryCode IN ('AFG', 'NLD');
```

```
SELECT * FROM Products  
WHERE Price BETWEEN 10 AND 20;
```

LIKE Operator & Wildcards

- LIKE searches for a pattern in a column.
- % matches zero or more characters; _ matches exactly one character.

SYNTAX

```
SELECT columns FROM table  
WHERE column LIKE pattern;
```

EXAMPLE

```
-- names starting with 'a'  
SELECT * FROM Customers  
WHERE CustomerName LIKE 'a%';  
  
-- countries containing 'island'  
SELECT Name FROM Country  
WHERE Name LIKE '%island%';
```

NULL Values & EXISTS

- A NULL field has no value — test it with IS NULL / IS NOT NULL, never with =.
- EXISTS returns true if a subquery returns one or more records.

SYNTAX

```
WHERE column IS NULL;
```

```
WHERE EXISTS  
  (SELECT col FROM t WHERE cond);
```

EXAMPLE

```
SELECT CustomerName FROM Customers  
WHERE Address IS NULL;
```

```
SELECT SupplierName FROM Suppliers s  
WHERE EXISTS (SELECT 1 FROM Products  
  WHERE SupplierID = s.SupplierID  
  AND Price < 20);
```

ORDER BY & LIMIT

- ORDER BY sorts the result-set — ascending by default, DESC for descending.
- LIMIT caps how many records are returned.

SYNTAX

```
SELECT columns FROM table  
ORDER BY col1 ASC|DESC  
LIMIT number;
```

EXAMPLE

```
-- top 10 cities by population  
SELECT * FROM City  
ORDER BY Population DESC  
LIMIT 10;
```

Aliases (AS)

- Aliases give a table or column a temporary, more readable name.
- The alias exists only for the duration of that query.

SYNTAX

```
SELECT column AS alias_name  
FROM table_name AS t;
```

EXAMPLE

```
SELECT CountryCode AS CC  
FROM City;  
  
SELECT CustomerID AS ID,  
       CustomerName AS Customer  
FROM Customers;
```

INSERT INTO Statement

- INSERT INTO adds new records to a table.
- Name the columns you supply; omitted columns take their DEFAULT (or NULL).

SYNTAX

```
INSERT INTO table_name  
  (column1, column2, ...)  
VALUES (value1, value2, ...);
```

EXAMPLE

```
INSERT INTO City (Name, CountryCode)  
VALUES ('Singapore', 'SGP');
```

UPDATE Statement

- UPDATE modifies existing records.
- ALWAYS include a WHERE clause — without it every row is updated.

SYNTAX

```
UPDATE table_name  
SET column1 = value1,  
    column2 = value2  
WHERE condition;
```

EXAMPLE

```
UPDATE City  
SET Name = 'Kabol'  
WHERE District = 'Kabol';
```

DELETE Statement

- DELETE removes existing records from a table.
- ALWAYS include a WHERE clause — without it every row is deleted.

SYNTAX

```
DELETE FROM table_name  
WHERE condition;
```

EXAMPLE

```
DELETE FROM City  
WHERE ID = 3285;
```

Hands-On Labs — Data Processing and Analysis

Lab 4

- Query Data with SELECT

Lab 5

- Filter with SQL Operators

Lab 6

- Insert, Update and Delete Records

Query Data with SELECT

LAB 4

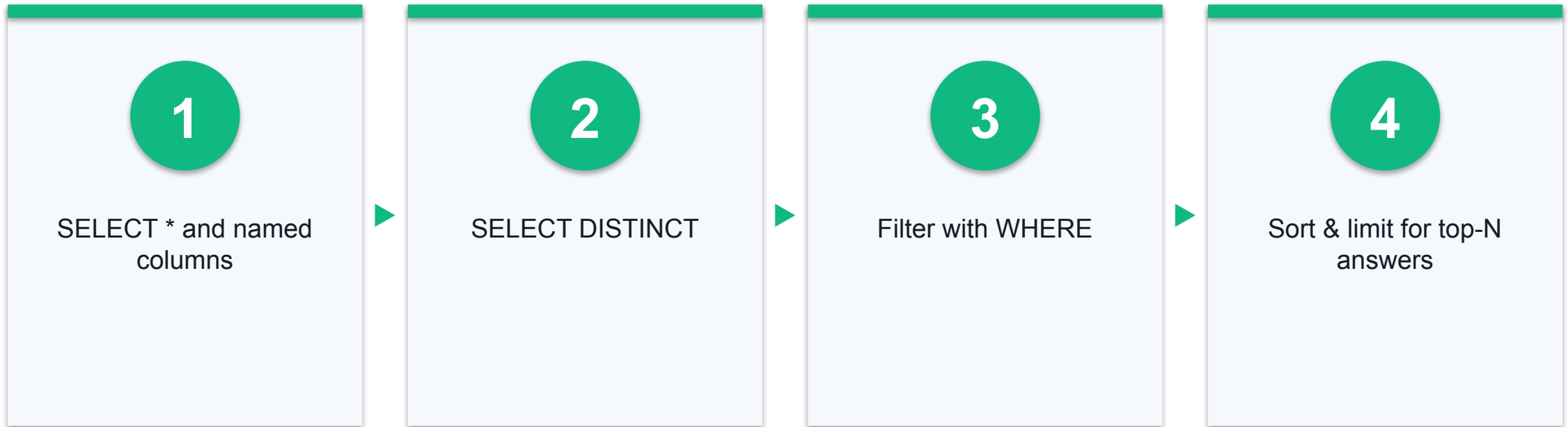
Query the world database with SELECT: retrieve all columns, chosen columns, distinct values and filtered rows, then combine sorting and limits to answer real questions.

You'll build

A set of working queries answering questions about world cities and countries, run against the world database.

Tools: SQLite Studio, world database

Lab 4 — How It Flows



Query Data with SELECT

Test it

Each query runs without error; the top-10 query returns 10 rows led by the world's most populous cities, and the Europe query returns 5 alphabetically-ordered European countries.

Filter with SQL Operators

LAB 5

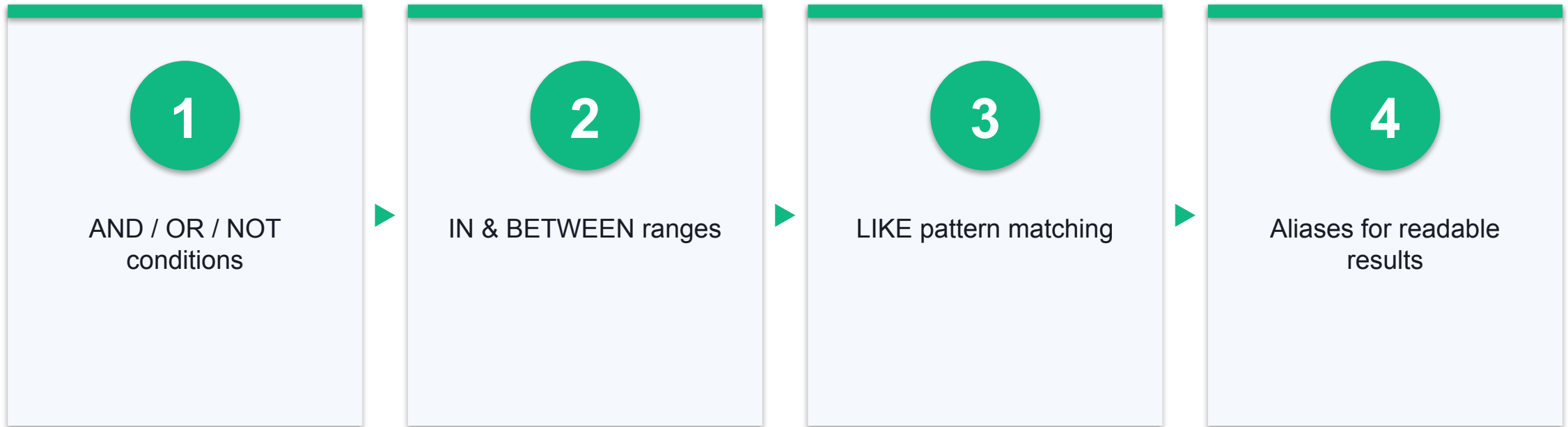
Sharpen your WHERE clauses with AND/OR/NOT, IN, BETWEEN, LIKE wildcards, NULL tests and aliases to slice the world data precisely.

You'll build

A query toolkit covering every SQL operator, answering pattern- and range-based questions on the world database.

Tools: SQLite Studio, world database

Lab 5 — How It Flows



Filter with SQL Operators

Test it

The IN query returns only AFG and NLD cities; the LIKE query lists island nations (Iceland, Marshall Islands, ...); the IS NULL query returns countries with an empty IndepYear, shown under the alias CountryName.

Insert, Update and Delete Records

LAB 6

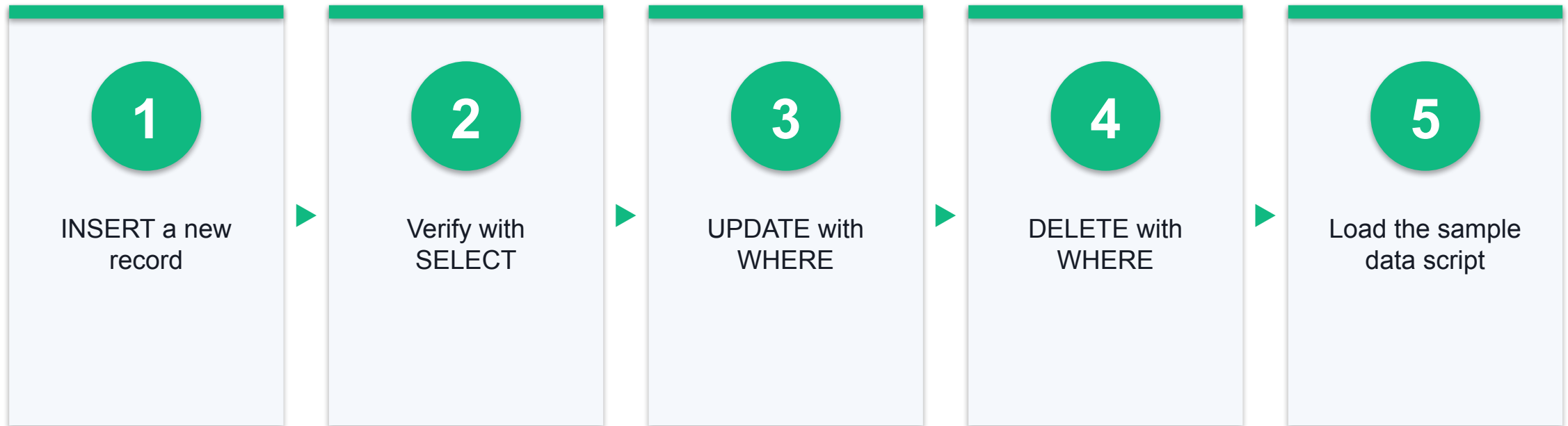
Change data safely: INSERT new city rows, UPDATE them with a WHERE clause, DELETE a specific record, and load the sample database from the provided insert script.

You'll build

A modified City table proving you can add, change and remove records — plus a fully-loaded sample database.

Tools: SQLite Studio, world & sample databases, insert script from the LMS

Lab 6 — How It Flows



Insert, Update and Delete Records

Test it

The Singapore row appears after the INSERT; the Kabul-district rows show the updated name; row ID 3285 is gone; and the three COUNT(*) queries return non-zero row counts in the sample database.

Recap — Data Processing and Analysis

- You can now: LO2 — apply data processing and analysis using SQL queries (A3, A5)
- You can now: LO2 — analyse data with operators, patterns and sorting (A3, A5)
- You can now: LO2 — process data by inserting, updating and deleting records (A3, A5)
- Full step-by-step instructions for every lab are in the Learner Guide.

Lunch Break

1:00 pm – 2:00 pm

TOPIC 03

Data Transformation

Aggregate Data · Join Data · Group Data

Key Concepts — Data Transformation

1

Aggregate functions

COUNT, AVG and SUM collapse many rows into one summary value.

2

Joins

INNER, LEFT, RIGHT and FULL OUTER joins combine rows from two tables on a matching key.

3

Set operations

UNION stacks the result-sets of two SELECTs with matching columns.

4

Grouping

GROUP BY aggregates per group; HAVING filters groups the way WHERE filters rows.

Aggregate Functions

1

COUNT()

Number of rows matching a criterion.

2

AVG()

Average value of a numeric column.

3

SUM()

Total of a numeric column.

4

MIN()

Smallest value in a column.

5

MAX()

Largest value in a column.

6

With WHERE

Aggregate only the rows that match a filter.

COUNT · AVG · SUM

- Aggregate functions collapse many rows into a single summary value.
- Combine with WHERE to summarise a filtered subset.

SYNTAX

```
SELECT COUNT(column) FROM t;  
SELECT AVG(column) FROM t;  
SELECT SUM(column) FROM t;
```

EXAMPLE

```
SELECT COUNT(CountryCode) FROM City;  
  
SELECT AVG(Population) FROM Country;  
  
SELECT SUM(Population) FROM Country;
```

SQL Joins — Four Ways to Combine Tables

1

INNER JOIN

Only rows with matching values in BOTH tables.

2

LEFT JOIN

All rows from the left table + matches from the right (NULL if none).

3

RIGHT JOIN

All rows from the right table + matches from the left (NULL if none).

4

FULL OUTER JOIN

All rows from both tables, matched where possible.

INNER JOIN Keyword

- INNER JOIN selects records that have matching values in both tables.
- The ON clause names the matching key columns.

SYNTAX

```
SELECT columns  
FROM table1  
INNER JOIN table2  
ON table1.col = table2.col;
```

EXAMPLE

```
SELECT City.Name, Country.Name  
FROM City  
INNER JOIN Country  
ON City.CountryCode = Country.Code;
```

LEFT JOIN & RIGHT JOIN

- LEFT JOIN returns ALL rows of the left table and matched rows of the right — NULL where there is no match.
- RIGHT JOIN mirrors this for the right table. (SQLite supports LEFT JOIN; emulate RIGHT by swapping the tables.)

SYNTAX

```
SELECT columns
FROM table1
LEFT JOIN table2
ON table1.col = table2.col;
```

EXAMPLE

```
SELECT c.CustomerName, o.OrderID
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
ORDER BY c.CustomerName;
```

FULL OUTER JOIN & UNION

- FULL OUTER JOIN returns all records when there is a match in either table.
- UNION combines the result-sets of two SELECTs — same column count, types and order. In SQLite, a FULL JOIN is emulated with LEFT JOIN + UNION.

SYNTAX

```
SELECT cols FROM t1  
FULL OUTER JOIN t2 ON t1.c = t2.c;
```

```
SELECT cols FROM t1  
UNION  
SELECT cols FROM t2;
```

EXAMPLE

```
SELECT l.*, r.* FROM left l  
LEFT JOIN right r ON l.id = r.id  
UNION  
SELECT l.*, r.* FROM right r  
LEFT JOIN left l ON l.id = r.id;
```

GROUP BY Statement

- GROUP BY collects rows sharing a value into summary rows — 'customers per country'.
- Used with COUNT, AVG, SUM, MIN, MAX to aggregate per group.

SYNTAX

```
SELECT col, AGG(col2)
FROM table
GROUP BY col
ORDER BY col;
```

EXAMPLE

```
SELECT CountryCode,
       AVG(Population)
FROM City
GROUP BY CountryCode;
```

HAVING Clause

- HAVING filters GROUPS — because WHERE cannot be used with aggregate functions.
- WHERE filters rows before grouping; HAVING filters the aggregated groups after.

SYNTAX

```
SELECT col, AGG(col2)
FROM table
GROUP BY col
HAVING AGG(col2) condition;
```

EXAMPLE

```
SELECT CountryCode, AVG(Population)
FROM City
GROUP BY CountryCode
HAVING AVG(Population) > 1000000;
```

Hands-On Labs — Data Transformation

Lab 7

- Aggregate Data with COUNT, AVG and SUM

Lab 8

- Join Data from Multiple Tables

Lab 9

- Group Data with GROUP BY and HAVING

Aggregate Data with COUNT, AVG and SUM

LAB 7

Summarise the world database with COUNT, AVG and SUM: count rows, average populations, total populations, and combine aggregates with DISTINCT and WHERE for real analytical questions.

You'll build

A summary report of world statistics — city counts, average and total populations, and filtered aggregate answers.

Tools: SQLite Studio, world database

Lab 7 — How It Flows



Aggregate Data with COUNT, AVG and SUM

Test it

Each aggregate returns a single-value result-set; the distinct head-of-state count is smaller than the country count, and the Europe query returns a small whole number.

Join Data from Multiple Tables

LAB 8

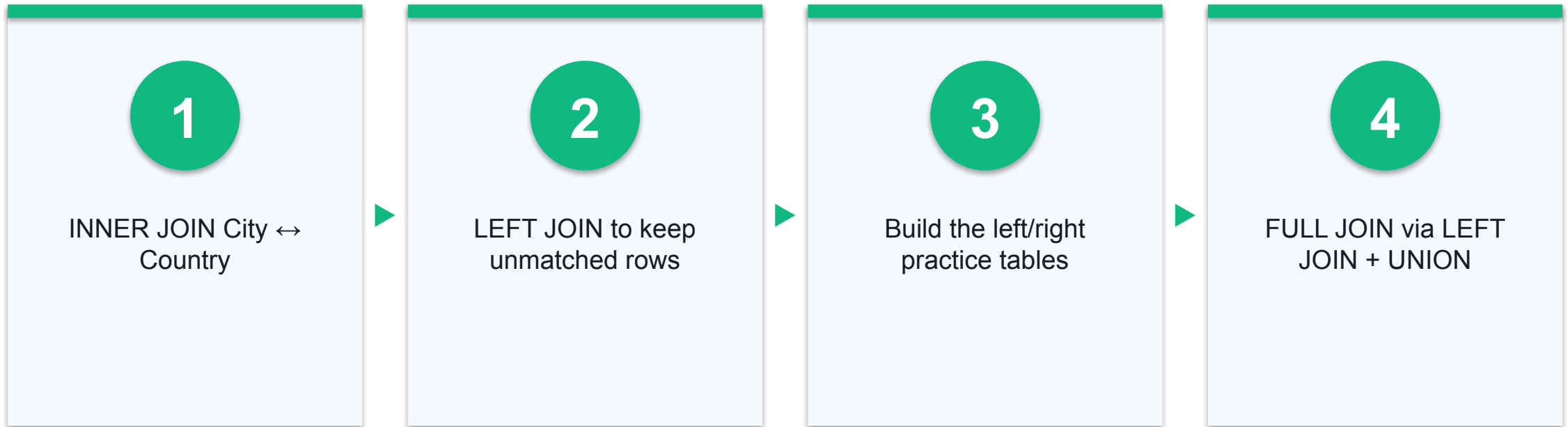
Combine tables: INNER JOIN City to Country on the country code, see how LEFT JOIN keeps unmatched rows, and emulate a FULL OUTER JOIN in SQLite using LEFT JOIN + UNION on a pair of practice tables.

You'll build

Joined result-sets linking cities to their countries, plus a full-outer-join of two practice tables showing matched and unmatched rows.

Tools: SQLite Studio, world & sample databases

Lab 8 — How It Flows



Join Data from Multiple Tables

Test it

The inner join returns only matched city–country rows; the LEFT JOIN lists every country (NULL capital where unmatched); and the full-join emulation returns ids 1–6 with NULLs on the sides that have no match.

Group Data with GROUP BY and HAVING

LAB 9

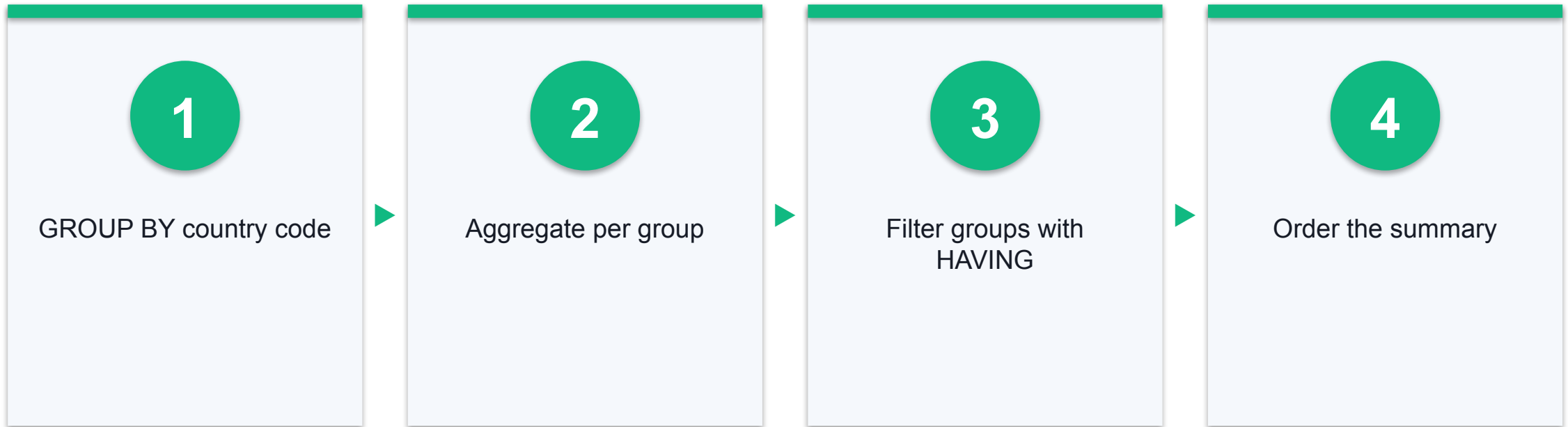
Aggregate per group: average city population by country code with GROUP BY, then keep only the populous groups with HAVING — and see why HAVING exists where WHERE cannot go.

You'll build

Grouped summaries of the world data — per-country averages, filtered to the groups that matter.

Tools: SQLite Studio, world database

Lab 9 — How It Flows



Group Data with GROUP BY and HAVING

Test it

The GROUP BY query returns one row per country code; the HAVING queries return only the qualifying groups (every AvgPop > 1,000,000; every Cities > 50), sorted descending.

Recap — Data Transformation

- You can now: LO3 — transform data with aggregate functions (A4, A6)
- You can now: LO3 — transform data from multiple sources with joins and unions (A4, A6)
- You can now: LO3 — transform data by grouping and filtering groups (A4, A6)
- Full step-by-step instructions for every lab are in the Learner Guide.

Tea Break

15 minutes

TOPIC 04

Introduction to Data Warehouse

Overview of Data Warehouse · Relational Databases · Stored Procedures

Key Concepts — Introduction to Data Warehouse

1

Data warehouse

A consolidated, historical, read-mostly database kept separate from operational systems for analysis.

2

OLAP vs OLTP

OLAP serves complex analytical queries; OLTP serves fast day-to-day transactions.

3

RDBMS & E-R model

The relational model plus entity-relationship design maps business entities to tables and keys.

4

Stored procedures

Prepared SQL saved on the server (SQL Server) for reuse — callable with parameters via EXEC.

What is a Data Warehouse?

1

Separate store

Kept apart from the organisation's operational database.

2

Historical

Consolidated historical data with no frequent updates.

3

For decisions

Helps executives organise, understand and use data strategically.

4

Integrating

Merges diverse application systems into one analytical view.

Data Warehouse vs Operational Database

Operational DB (OLTP)

- Built for known tasks: search, index, update
 - Many concurrent short transactions
 - Current, constantly-changing data
 - Normalised for write efficiency

Data Warehouse (OLAP)

- Built for complex analytical queries
 - Read-mostly, batch loaded
 - Consolidated historical data
 - Structured for fast analysis

Data Warehouse Features

1

Subject oriented

Organised around subjects — product, customers, sales — not daily operations.

2

Integrated

Built by integrating data from heterogeneous sources into one schema.

3

Time variant

Data is identified with a time period, enabling trend analysis.

4

Non-volatile

Loaded data is not erased when new data is added.

Data Warehouse Applications

1

Financial services

Portfolio and risk analysis.

2

Banking

Customer and product profitability.

3

Consumer goods

Sales and promotion analysis.

4

Retail

Basket analysis and inventory.

5

Manufacturing

Controlled process optimisation.

6

Closed loop

Plan → execute → assess
feedback for management.

OLAP Servers

1

ROLAP

Relational OLAP — analysis directly over relational tables.

2

MOLAP

Multidimensional OLAP — pre-computed data cubes.

3

HOLAP

Hybrid OLAP — relational detail with cube summaries.

4

Specialized SQL Servers

Engines tuned for analytical SQL workloads.

RDBMS — the Basis for SQL

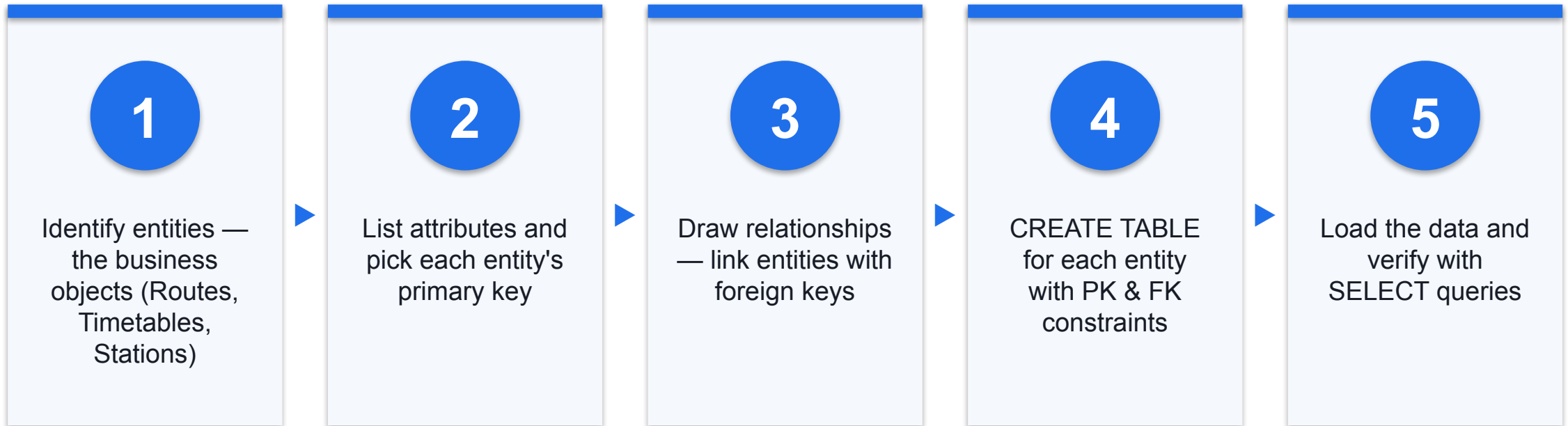
What an RDBMS is

- Relational Database Management System
 - Data lives in tables of rows and columns
 - Basis of SQL Server, DB2, Oracle, MySQL, Access
 - Tables split into fields (columns) and records (rows)

Why it wins

- Simple yet powerful relational model
 - Tracks inventories, e-commerce, customer data
 - Fits any data where points relate to each other
 - Managed securely, consistently, at scale

E-R Model → Database



Case Study: SMRT Public Transport

The business

- SMRT operates buses, taxis and trains across Singapore
 - Needs schedules, routes and disruption history
 - Analyses service impact and severity

The E-R model

- Routes — route_id (PK), type, code, name, direction
 - Timetables — timetable_id (PK), route_id (FK), station_code (FK), frequency, first/last trip
 - DisruptedRoutes — disrupt_no (PK), date, type, details, timetable_id (FK)

Stored Procedures (SQL Server)

- A stored procedure is prepared SQL saved on the server, reusable by name.
- Not available in SQLite — this is SQL Server / MySQL territory.

SYNTAX

```
CREATE PROCEDURE procedure_name  
AS  
sql_statement  
GO;  
  
EXEC procedure_name;
```

EXAMPLE

```
CREATE PROCEDURE SelectAllCustomers  
AS  
SELECT * FROM Customers  
GO;  
  
EXEC SelectAllCustomers;
```

Stored Procedure with Parameters

- Parameters let one procedure act on the values passed to it.
- Call EXEC with the parameter values.

SYNTAX

```
CREATE PROCEDURE name @param type
AS
sql_statement_using @param
GO;

EXEC name @param = value;
```

EXAMPLE

```
CREATE PROCEDURE SelectAllCustomers
    @City nvarchar(30)
AS
SELECT * FROM Customers
WHERE City = @City
GO;

EXEC SelectAllCustomers
    @City = 'London';
```

Hands-On Labs — Introduction to Data Warehouse

Lab 10

- Map an E-R Model to Database Tables (SMRT Case Study)

Lab 11

- Author Stored Procedures for SQL Server

Map an E-R Model to Database Tables (SMRT Case Study)

LAB 10

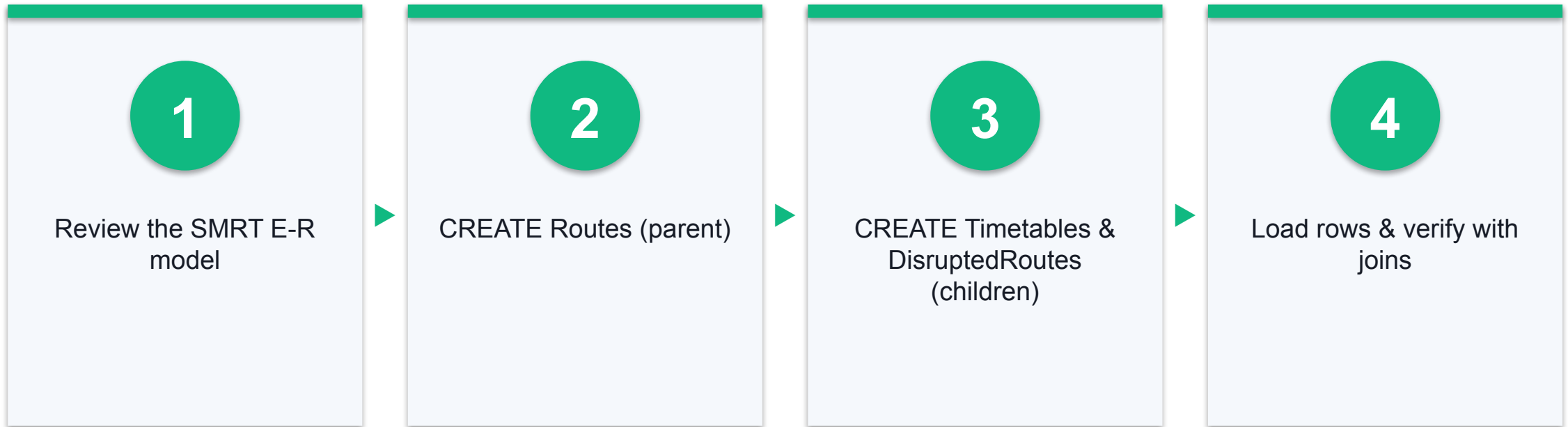
Take the SMRT public-transport E-R model (Routes, Timetables, DisruptedRoutes) and map it to physical tables: create each entity with its primary key, wire the foreign-key relationships, and verify the model with joined queries.

You'll build

A three-table transport schema in SQLite that faithfully implements the SMRT E-R model with PK/FK mappings.

Tools: SQLite Studio, sample database

Lab 10 — How It Flows



Map an E-R Model to Database Tables (SMRT Case Study)

Test it

The three tables exist with their PK/FK constraints visible in the DDL, and the three-way join returns one row linking 'North South Line' → 'NS01' → 'Signal fault'.

Author Stored Procedures for SQL Server

LAB 11

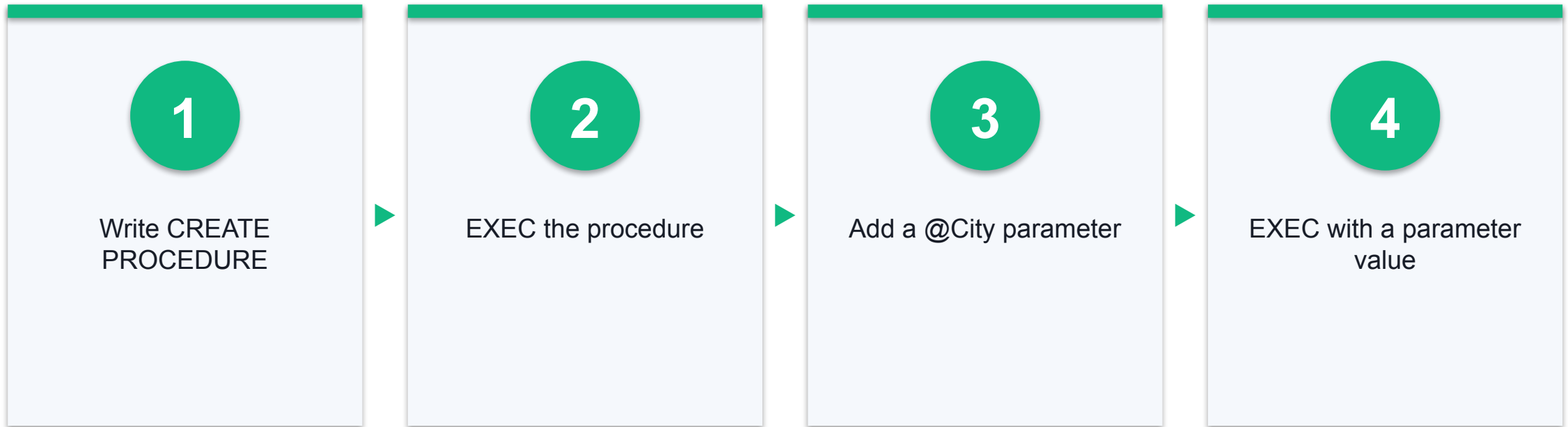
Write stored procedures in SQL Server syntax — a plain procedure and a parameterised one — and understand where they run (SQL Server / MySQL, not SQLite) and why warehouses use them for repeatable batch work.

You'll build

Two stored-procedure scripts (plain and parameterised) ready to run on SQL Server, plus the EXEC calls that invoke them.

Tools: SQL Editor, SQL Server syntax (sqliteonline.com's MS SQL engine, or any SQL Server)

Lab 11 — How It Flows



Author Stored Procedures for SQL Server

Test it

Both procedures create without error; EXEC SelectAllCustomers returns the full customer list, and EXEC SelectCustomersByCity @City = 'London' returns only the London rows.

Recap — Introduction to Data Warehouse

- You can now: LO4 — apply data mapping from an E-R model to a data warehouse schema (A7, A8, A9)
- You can now: LO4 — operate data-warehouse platform capabilities with reusable stored procedures (A7, A8, A9)
- Full step-by-step instructions for every lab are in the Learner Guide.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Data Modeling

Created databases, tables, constraints, keys and relationships in SQLite Studio.

2

Data Processing & Analysis

Queried, filtered, sorted, inserted, updated and deleted data with SQL.

3

Data Transformation

Aggregated, joined, unioned and grouped data across multiple tables.

4

Data Warehouse

Mapped an E-R model to tables and authored stored procedures.

Recommended Next Courses

1

Building Professional Websites with WordPress

WSQ · Tertiary Infotech

2

Building a Successful eCommerce Store with WooCommerce

WSQ · Tertiary Infotech

3

Mastering HTML5 and CSS3 Fundamentals for Professional Web Design

WSQ · Tertiary Infotech

4

Hands-On Web App Development with Javascript

WSQ · Tertiary Infotech

5

Running a Successful eCommerce Store with Shopify

WSQ · Tertiary Infotech

Support

1

Email

enquiry@tertiaryinfotech.com

2

Telephone

+65 6100 0613

3

Website

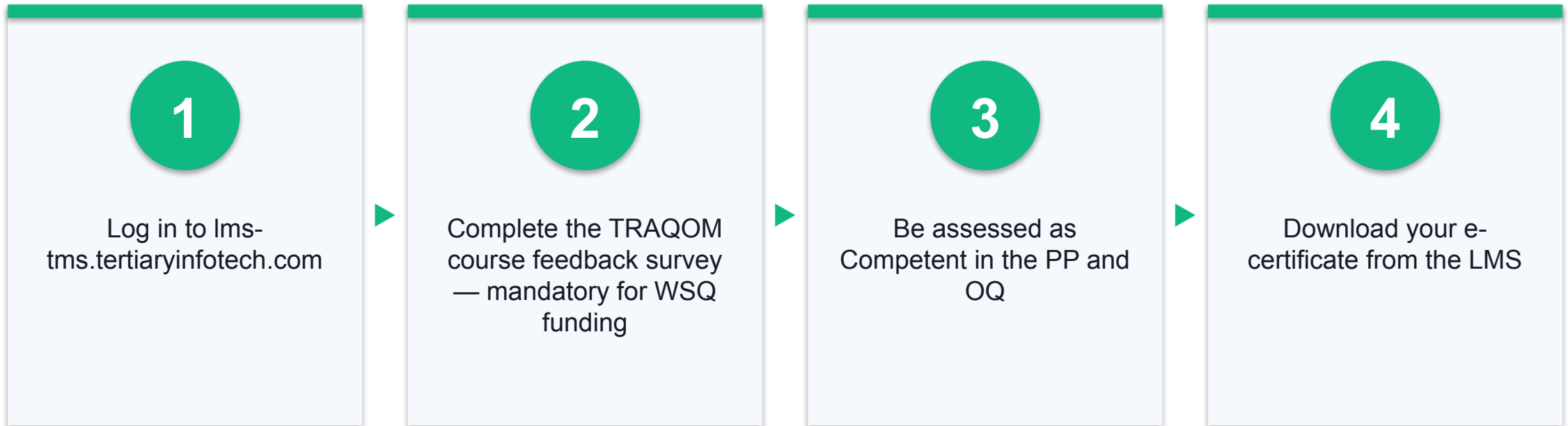
www.tertiarycourses.com.sg

4

LMS / TMS

lms-tms.tertiaryinfotech.com

Certificate & TRAQOM Survey (Mandatory)



Final Assessment

1

Practical Performance (PP)

Hands-on SQL tasks · 70 minutes · open book.

2

Oral Questioning (OQ)

Spoken knowledge questions · 20 minutes · one-to-one.

3

Before you begin

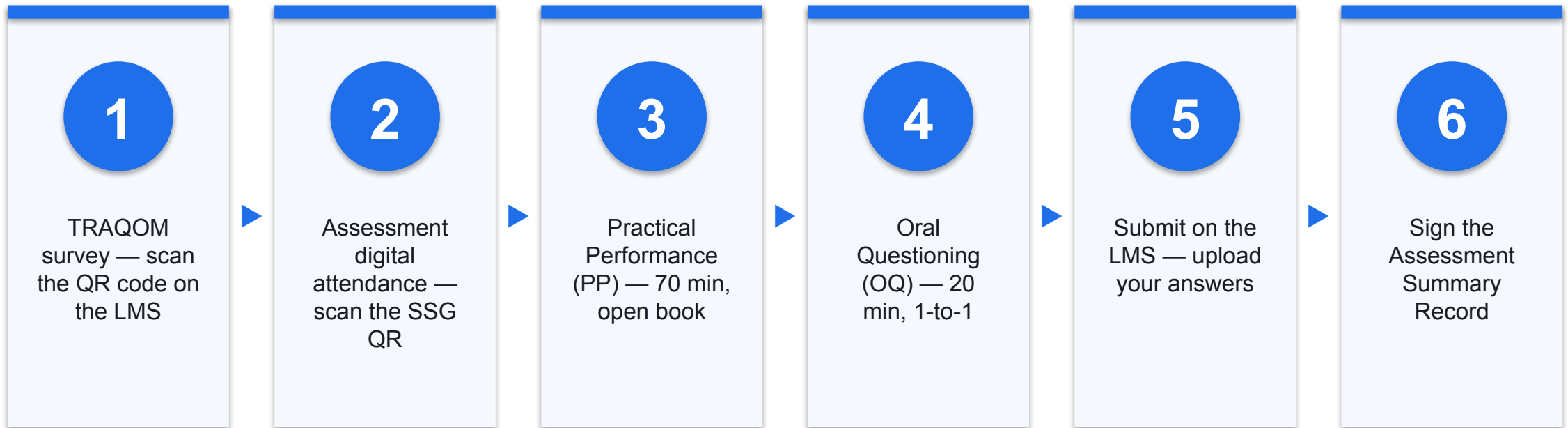
Take the Assessment digital attendance (TRAQOM).

4

After the OQ

Submit your work on the LMS, then sign the Assessment Summary Record.

Assessment Flow



Digital Attendance (Mandatory)

1

Three times a day

Take the AM, PM and Assessment digital attendance — mandatory for WSQ-funded courses.

2

QR code from SSG

The trainer or administrator displays the attendance QR code from the SSG portal.

3

Scan and submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

A minimum 75% attendance rate is required to be eligible for assessment and funding.

SEE YOU IN THE NEXT COURSE

Thank You!

You can now model, query, transform and warehouse data with SQL — happy querying!